

Bug Fixes Report - CoWork Booking API

Date: July 9, 2026 | **Author:** Antigravity AI | **Status:** Completed & Verified

This report documents the resolution of all identified software bugs inside the CoWork Booking API.

All changes are verified using automated unit/integration tests.

1. Booking Reference Code Race Condition (reference.py)

Bug: Monotonic reference code generation was unsynchronized, causing race conditions where concurrent creation requests fetched duplicate values.

Fix: Introduced a `threading.Lock`` to synchronize counter updates.

BEFORE APP/SERVICES/REFERENCE.PY

```
def next_reference_code() -> str: current = _counter["value"] _format_pause()  
_counter["value"] = current + 1 return f"CW-{current:06d}"
```

AFTER APP/SERVICES/REFERENCE.PY

```
def next_reference_code() -> str: with _lock: current = _counter["value"] _format_pause()  
_counter["value"] = current + 1 return f"CW-{current:06d}"
```

2. Python 3.12+ `datetime.utcnow()` Deprecation warnings

Bug: Multiple warning flags were raised by SQLAlchemy model defaults and endpoint datetimes due to deprecated `datetime.utcnow()` usage.

Fix: Replaced all instances of `datetime.utcnow()` with timezone-aware `datetime.now(datetime.timezone.utc).replace(tzinfo=None)`.

3. Deadlock in Booking Lifecycle Notifications (notifications.py)

Bug: ``notify_created`` nested ``_email_lock`` then ``_audit_lock``, whereas ``notify_cancelled`` nested ``_audit_lock`` then ``_email_lock``. Concurrent lifecycles caused threads to lock up indefinitely.

Fix: Replaced nested locking with sequential lock acquisitions.

BEFORE APP/SERVICES/NOTIFICATIONS.PY

```
def notify_created(booking) -> None: with _email_lock: _send_email("created", booking) with  
_audit_lock: _write_audit("created", booking)
```

AFTER APP/SERVICES/NOTIFICATIONS.PY

```
def notify_created(booking) -> None: with _email_lock: _send_email("created", booking) with  
_audit_lock: _write_audit("created", booking)
```

4. Datetime Parsing & Timezone Normalization (timeutils.py)

Bug: Timezone-aware datetimes had their offsets stripped (via `replace(tzinfo=None)`) without converting the actual hour/minute values to UTC first.

Fix: Normalizes the datetime object to UTC using `astimezone(timezone.utc)` before removing the timezone info.

5. Booking Time Validations (bookings.py)

Bugs: 1. Allowing past bookings up to 5 minutes ago (grace window). 2. Failing to reject bookings where `end_time ≤ start_time` (negative or zero durations).

Fixes: 1. Changed past-time check to strictly reject start times in the past (`start ≤ now`).
2. Added a validation to reject requests where `end ≤ start`.

6. Back-to-Back Overlap Conflict Check (bookings.py)

Bug: Back-to-back bookings (one starting exactly when another ends) were flagged as conflicts because the check used inclusive operators (`≤`).

Fix: Replaced the comparison operator with strict inequality (<).

7. Booking Pagination, Offset, and Sorting (bookings.py)

Bugs: 1. Used wrong offset (skipped the first page because it used `page * limit` instead of `(page - 1) * limit`). 2. Sorted descending instead of chronologically (ascending by `start_time`). 3. Ignored the requested query limit (hardcoded `.limit(10)`).

Fixes: Corrected the query offsets, limit variables, and sorting order.

8. Visibility Leak & Detail Redundancy (bookings.py)

Bugs: 1. Members could view details of other organization members' bookings. 2.

Overwrote the `start_time` property in the single booking detail response.

Fixes: 1. Implemented a non-admin ownership restriction checking ``booking.user_id == user.id``. 2. Removed the buggy response `["start_time"] = iso_utc(booking.created_at)` line.

9. Cancellation Refund Policies & Precision Rounding (bookings.py & refunds.py)

Bugs: 1. Notice under 24 hours incorrectly yielded a 50% refund (should be 0%). 2. Notice of exactly 48 hours incorrectly yielded a 50% refund (should be 100%). 3. Rounding of refund amounts used integer casts (`int(refund_dollars * 100)`), causing truncation of cents.

Fixes: 1. Corrected the notice calculation conditional branches in `bookings.py`. 2. Changed the rounding formula in `refunds.py` to use Python's `round()` function on whole cents.

10. Reporting & Availability Caching (bookings.py)

Bugs: 1. Availability cache was not invalidated when a booking was cancelled. 2. Usage report cache was not invalidated when a booking was created.

Fixes: 1. Added `cache.invalidate_availability()` to the cancellation endpoint. 2.

Added `cache.invalidate_report()` to the creation endpoint.

11. Auth Endpoint Issues (auth.py & routers/auth.py)

Bugs: 1. Token lifetime evaluated to 15 hours instead of 15 minutes. 2. Token revocation evaluated user IDs (sub) instead of token IDs (jti). 3. Refresh tokens were reusable, causing security issues. 4. Registration allowed duplicate usernames to bypass credentials.

Fixes: Corrected lifetime math, swapped checked claim to `jti`, implemented refresh revocation on refresh, and threw 409 USERNAME_TAKEN.

12. Concurrency & Thread Safety (ratelimit.py & stats.py)

Bugs: 1. Accessing and filtering rate-limiting rolling window buckets concurrently caused race-prone behavior. 2. Increments and decrements of live room booking stats concurrently overwrote the stats dictionary.

Fixes: Used Python's `threading.Lock` to synchronize all accesses.

13. Admin Export Organizational Scoping Bypass (export.py)

Bug: Admin export route fetched raw bookings for a room without verifying it belonged to the caller's organization.

Fix: Enforces organization ownership check on the `room_id`, raising 404 `ROOM_NOT_FOUND` if invalid.

14. Test Suite Verification

All unit and integration tests successfully pass with only Starlette library warnings remaining:

```
$ pytest
tests/test_auth.py .... [ 21%]
tests/test_bookings.py ..... [ 68%]
tests/test_smoke.py . [ 73%]
tests/test_timeutils.py ..... [100%]
```

```
===== 19 passed, 1 warning in 13.09s =====
```